

AMOL KUMAR THAKRE

PHD, STATISTICAL PHYSICS — SCIENTIST, RESEARCHER & EDUCATOR

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Research & Teaching Portfolio (web)

PROFILE

PhD scientist and educator bridging **molecular fluid dynamics and industrial-scale engineering**, with research tenure across the Netherlands, Norway, and India and **10+ years of industrial R&D**. Author of a **doctoral monograph**, **four first-author journal papers**, and the *NEET Preparation Companion* (Amazon, 2026). Now applies research-grade rigour to elite exam preparation — **concept-first coaching for JEE, NEET, and Board examinations** in small, personally mentored batches.

TEACHING & MENTORSHIP

Academic Coach — Physics, Chemistry, Mathematics *JEE/NEET/Boards* Bangalore **2021 – Present**

- Mentors **15+ students annually** in small batches of 10 – 15, across three structured pathways: **Foundation** (Class 9 – 10; NCERT with deeper exploration, Board + Olympiad), **Two-Year** (Class 11 – 12, JEE / NEET), and a high-intensity **One-Year** dropper track.
- Methodology — first principles before formulae**: every concept traced back to fundamental laws; deep understanding first, exam technique second.
- Scientific error tracking**: every mock test mapped with chapter-wise analytics, separating concept gaps from execution errors; bi-weekly to weekly full-length mocks with one-on-one tracking.
- Vedic speed & agility**: 16 sutras of Vedic mathematics integrated for mental calculation speed.
- Built **web-based adaptive question banks** and AI-assisted revision tools (Claude, Gemini) for student-paced, targeted practice.

Simulation Instructor *Graduate MD/DPD Laboratories* University of Twente **2004 – 2006**

- Designed and delivered graduate simulation laboratories: theory, code architecture, execution, and scientific interpretation; scaffolded students from first principles to peer-review-quality analysis.

Research Supervisor *Master's Projects, Rig-Data Analytics* Equinor, Norway **2011 – 2013**

- Defined research scope, mentored, and evaluated Master's theses; introduced Python scripting and version-controlled collaborative workflows.

BOOKS

- NEET Preparation Companion* — Amazon, **2026** (amazon.in/dp/B0H469W264). A structured, concept-first guide distilling years of classroom insight into a preparation framework for NEET aspirants.
- JEE Preparation Companion* — forthcoming, **2026**. The same proven methodology tailored for JEE Main & Advanced — conceptual depth and exam precision built in parallel.
- Doctoral monograph**: *Simulation of phase separation in quiescent and sheared liquids*. University of Twente Press, **2008**. ISBN 978-90-365-2719-4.

JOURNAL PUBLICATIONS

- Domain formation and growth in spinodal decomposition of a binary fluid by molecular dynamics simulations. *Physical Review E* **77**, 011503 (2008).
- MD simulations of phase-separating binary liquids in cylindrical Couette flow. *Journal of Chemical Physics* **128**, 154707 (2008).
- Finite system-size effects in the interfacial dynamics of binary liquid films. *Journal of Chemical Physics* **129**, 044701 (2008).
- Spinodal decomposition of asymmetric binary fluids in a micro-Couette geometry simulated with molecular dynamics. *Journal of Chemical Physics* **129**, 074505 (2008).
- Heat penetration in multi-phase liquids and effect of controlled shaking. *Submitted, Journal of Food Science*.

RESEARCH EXPERIENCE

University of Twente *Doctoral Researcher, Statistical Physics* Netherlands **2003 – 2008**

- Thesis *Phase separation in complex fluids: A molecular view point* — supervisors Prof. W. K. den Otter and Prof. W. J. Briels (thesis PDF).

- Built bespoke **3D parallelised MD and DPD codes** (C / FORTRAN); spinodal decomposition, interfacial dynamics, and Couette-flow instabilities in binary fluids.

Industrial Research *Equinor; GE / BHGE; SINTEF; Nofima*

Norway & India **2008 – 2018**

- **Equinor (2009 – 13)**: designed and supervised large-scale heavy-oil rig experiments; field-facing design recommendations for sub-sea separators and electrostatic coalescers (Peregrino, Bressay).
- **GE / BHGE (2013 – 18)**: Digital-Twin slug-risk prediction from high-fidelity CFD; $10\times$ O₂ mass-transfer gain from bioreactor redesign; supersonic steam-turbine cascade aerodynamics.
- **SINTEF & Nofima (2008 – 09)**: thermal intensification via microwave and RF heating; computational transport models for industrial food processing; separator design.
- Continues as independent technical consultant (2018 – present): scientific computing, multi-phase flow, and AI / ML modelling for global energy and manufacturing clients.

EDUCATION

PhD, Statistical Physics — University of Twente, Netherlands

2003 – 2008

Molecular simulation of phase separation in complex fluids.

M.S., Chemical Engineering — Indian Institute of Science (IISc), Bangalore

2001 – 2003

Thesis: *Phase inversion in liquid-liquid dispersions*.

B.E., Chemical Engineering — Madhav Institute of Technology and Science, Gwalior

1996 – 2000

Economics and technology of chemical processes and plant design.

Executive & Professional — ISB Hyderabad; MIT

2006 – 2019

ISB — Leading Digital Business Transformation & Innovation (2018 – 19); MIT — Complex Physical, Biological & Social Systems; Networks, Modeling & Evolution (2006); The Analytics Edge (online, 2016).

TEACHING PHILOSOPHY

Industrial R&D logic applied to elite exam preparation, on three pillars: **first principles** — every concept traced back to fundamental laws, *why* before *what*; **deep understanding** — reasoning over regurgitation; and **exam precision** — once understanding is in place, technique becomes natural. Fuses mental-agility training (Vedic mathematics) with data-driven error analytics from industrial research practice.

TALKS & MEDIA

- **Digital Twin Strategy** — recorded technical talk (YouTube).
- **Technology Showcase** — research & engineering portfolio presentation (YouTube).
- Doctoral thesis openly available via the University of Twente research repository (PDF).

RESEARCH INTERESTS & AFFILIATIONS

Interests: CFD, statistical physics, MD / DPD, multi-phase flows, phase transitions, stochastic modelling, AI / ML in the physical sciences, physics and mathematics pedagogy.

Affiliations: Member, JMB Centre of Fluid Dynamics, Netherlands · Associate, New England Complex Systems Institute (NECSI), Boston, USA · Member, Chemical Engineers Association of India.